

Chapter 2

Computer Organization and Architecture

2.1 INTRODUCTION:

- **Computer Architecture** is a blueprint for design and implementation of a computer system. It refers to the overall design of a computer system, including the hardware and software components that make up the system and how they interact with each other.
- **Computer Organization** refers to the way in which the hardware components of a computer system are arranged and interconnected.
- Computer Architecture explains what a **computer should do**, whereas Computer Organization explains **how a computer works**.
- Example as Car: Computer architecture defines the what (what features the car has, like engine type, seating capacity, etc.), while computer organization defines the how (how those features are physically implemented and connected).

2.2 Central Processing Unit

- The central processing unit (CPU) is referred to as the —brain of a computer system and it converts **data (input) into meaningful information (output)**.
- It controls all internal and external devices, performs arithmetic and logic operations, and operates only on binary data, that is, data composed of 1s and 0s.
- Eventually, the results of these operations are translated into characters, numbers and symbols that can be understood by the user.
- In addition, it also controls the usage of the main memory to store data and instructions, and controls the sequence of operations.

2.2.1 Arithmetic/Logic Unit

- The arithmetic/logic unit (ALU) contains the electronic circuitry that executes all arithmetic and logical operations on the data made available to it. The data required to perform the arithmetic and logical functions are inputs from the designated registers.

The ALU comprises two units: Arithmetic Unit (AU) and Logic Unit (LU).

Arithmetic Unit:

- The AU contains the circuitry that is responsible for performing the actual computing and carrying out arithmetic calculations, such as addition, subtraction, multiplication, and division. It can perform these operations at a very high speed.

Logic Unit:

- These operations are logical comparisons between data items.
- The LU can compare numbers, letters or special characters and then take action based on the result of the comparison.

Logical operations of the LU test for mainly three conditions:

- **Equal-to Condition:** In a test for this condition, the ALU compares two values to determine if they are equal. For example, if the number of tickets sold equals the number of seats in a cinema hall, then houseful is declared.
- **Less-than Condition:** To test this condition, the ALU compares values to determine if one is less than the other; for example, if a person purchases less than two tickets, then no discount rate is given.
- **Greater-than Condition:** In this type of comparison, the computer determines if one value is greater than the other; for example, if a person purchases greater than two tickets, then the discount rate is 5 per cent.

2.2.2 Registers

- The registers are special-purpose, high-speed temporary memory units.
- **Registers** hold operands or instructions that CPU would be currently processing such as data, instructions, and intermediate results of calculations
- The registers can be considered as the CPU's working memory. **Memory** holds instructions and the data about the currently executing program required by the CPU.
- As soon as a particular instruction or piece of data is processed, the next instruction immediately replaces it, and the information that results from the processing is returned to the main memory.

Table 2.1 Registers and Functions

Register Name	Function
Program Counter (PC)	The PC keeps track of the next instruction to be executed
Instruction Register (IR)	The IR holds the instruction to be decoded by the CU
Memory Address Register (MAR)	The MAR holds the address of the next location in the memory to be accessed
Memory Buffer Register (MBR)	The MBR is used for storing data either coming to the CPU or data being transferred by the CPU
Accumulator (ACC)	The ACC is a general-purpose register used for storing temporary results and results produced by the ALU
Data Register (DR)	The DR is used for storing the operands and other data

2.2.3 Control Unit

- The control unit (CU) of the CPU contains circuitry that uses electrical signals to direct the entire computer system to carry out, or execute, stored program instructions.

- The CU also does not execute program instructions; rather, it directs other parts of the system to do so by communicating with both the ALU and the memory.
- The CU controls the I/O devices and transfer of data to and from the primary storage.
- Instructions are retrieved from the primary storage, one at a time.
- For this, the CU uses the IR(Instruction Register) for holding the current instruction and an instruction pointer to hold the address of the next instruction.

2.2.4 System Bus

- A **bus** is a set of connections between two or more components/devices, which are designed to transfer several/all bits of a word (is a collection of these bits, such as 16, 32, or 64 bits) from the source to the destination.
- A bus consists of multiple paths, which are also termed lines, and each line is capable of transferring one bit at a time. Thus, to transmit 8 bits simultaneously over a bus, eight lines are required.
- A bus can be unidirectional (transmission of data can be in only one direction) or bi-directional (transmission of data can be in both directions). In a shared bus, only one source can transmit data at one time, while one or more than one can receive that signal.
- A bus that connects to all three components (**CPU, memory, I/O devices**) is called a **system bus**.

System Bus - Functional groups

Address Lines/bus:

- A collection of wires used to identify particular location in main memory is called Address Bus
- As the memory may be divided into a linear array of bytes or words, for reading or writing any information on to the memory, the CPU needs to specify the address of a particular location.
- **Example:**
A system with 4-bit address bus can address $2^4 = 16$ Bytes of memory.
A system with 16-bit address bus can address $2^{16} = 64$ KB of memory
A system with 20-bit address bus can address $2^{20} = 1$ MB of memory

Data Lines/Bus:

- A collection of wires through which data is transmitted from one part of a computer to another is called Data Bus. Normally, a data bus consists of 8, 16 or 32 separate lines.
- The data bus is bidirectional because the data can flow in either direction from CPU to memory (or input/output device) or from memory to the CPU.
- The number of lines present in a data bus is called the width of the data bus.
- **Example:**

A 16-bit bus can transmit 16 bits of data at a time.

32-bit bus can transmit 32 bits at a time.

Control Lines/Bus:

- Control lines are used to control the access to data and the address bus; The control lines are collectively called a control bus
- These lines are used for transmission of commands and timing signals (which validate data and addresses) between the system modules.
- Timing signals indicate whether data and address information is valid, whereas command signals specify which operations are to be performed.

2.2.5 Main Memory Unit

- The memory is that part of the computer that holds data and instructions for processing.
- Logically, it is an integral component of the CPU, but, physically, it is a separate part placed on the computer's motherboard.
- The memory stores program instructions or data for only as long as the program they pertain to is in operation.
- The CPU accesses the main memory in a random manner, that is, the CPU can access any location of this memory to either read information from it or store information in it.
- The primary memory is of two types: the first is called the **Random Access Memory (RAM)** and the second is the **Read Only Memory (ROM)**.

Random Access Memory (RAM)

- It is one of the parts of the Main memory, also famously known as Read Write Memory.
- It can be defined as a block of sequential memory locations, each of which has a unique address determining the location and those locations contain a data element. Storage locations in the main memory are addressed directly by the CPU's instructions.
- It is **volatile in nature**, which means the information stored in it remains as long as the power is switched on. As soon as the power is switched off, the information contained in the RAM is lost.
- RAM can help in both reading and writing. (To better understand RAM, imagine the blackboard of the classroom, the students can both read and write and also erase the data written after the class is over, some new data can be entered now)

Read Only Memory (ROM)

- The ROM stores the initial start-up process of the computer and routines in the Basic Input/Output System (BIOS – Basic Input Output System) which can only be read by the CPU, each time it is switched on.

- The contents of the ROM are not lost even in the case of a sudden power failure, thus making it **non-volatile in nature**.
- The ROM is also randomly accessible in nature, which means the CPU can randomly access any location within the ROM. Improvement in technology for constructing a flexible ROM comes in various types, namely Programmable **Read Only Memory (PROM)**, Erasable Programmable Read Only Memory, Erasable Programmable Read only Memory (EPROM) and Electrically Erasable Programmable Read only Memory (EEPROM).

Difference between RAM and ROM

RAM	ROM
RAM is a volatile memory that could store the data as long as the power is supplied.	ROM is a non-volatile memory that the could retain the data even when the power is turned off.
Read and write operations are supported.	Only read operations are supported.
It is a high-speed memory	It is much slower than the RAM.
CPU can easily access data stored in RAM.	CPU cannot easily access data stored in ROM.
Used to store the data that has to be currently processed by CPU temporarily.	It is typically used to store firmware or microcode, which is used to initialize and control hardware components of the computer.

2.2.6 Cache Memory

- The cache is a very high speed, expensive piece of memory, which is used to speed up the memory-retrieval process.
- The most important use of cache memory is that it is used to reduce the average time to access data from the main memory.
- Without the cache memory, every request from the CPU to access data would be sent to the main memory, which in response sends the data back to the CPU through the system bus. This is a slow process in computing terms.
- The idea of introducing cache is that this extremely fast memory would store data that are frequently accessed.

L1 Cache:

- This cache is closest to the processor and hence is termed as the primary or L1 cache.

- In a typical CPU, the primary cache ranges in size from 8 to 64 KB, with larger amounts on the newer processors.

L2 Cache:

- The L2 cache is larger but slower in speed than L1 cache. It is used to see recent accesses that is not picked by the L1 cache and is usually 64 to 2 MB in size.
- The L2 cache is also found on the CPU.
- If the L1 and L2 cache are used together, then the missing information that is not present in the L1 cache can be retrieved quickly from the L2 cache.

L3 Cache:

- It is an extra cache built into the motherboard between the processor and main memory to speed up the processing operations.
- It reduces the time gap between the request and retrieval of the data and instructions, thereby accessing data much more quickly than the main memory.
- The L3 cache is being used with processors nowadays having more than 3 MB of storage in it.